

International Institute of Information Technology, Bangalore
CSE 102P Data Structures and Algorithms Lab- Sec A
Test1: 4 February 2026.

You are given four positive integers a, b, c, d (with $a \leq b$ and $c \leq d$). Define:

- S_1 = set of all even numbers in the range $[a, b]$ (inclusive)
- S_2 = set of all multiples of 3 in the range $[c, d]$ (inclusive)

Let $U = S_1 \cup S_2$ (union of S_1 and S_2). The **rank** of an element x in set S is defined as:

$$\text{rank}(x) = 1 + (\text{number of elements in } S \text{ that are } > x)$$

In other words, elements are ranked in **descending order** (largest element has rank 1, second largest has rank 2, etc.).

You are also given an integer k . Your task is to find the element in U that has rank k . If no such element exists (i.e., $k > |U|$ or $k < 1$), print -1 .

Input Format

A single line containing five space-separated integers: $a \ b \ c \ d \ k$

Output Format

A single integer: the element with rank k in U , or -1 if no such element exists.

Examples

Input	Output
1 10 5 15 3	10
1 5 6 10 1	9
1 1 1 1 1	-1
2 100 3 100 10	94

Explanation of Examples

Example 1: $a = 1, b = 10, c = 5, d = 15, k = 3$

- S_1 = even numbers in $[1, 10] = \{2, 4, 6, 8, 10\}$
- S_2 = multiples of 3 in $[5, 15] = \{6, 9, 12, 15\}$
- $U = S_1 \cup S_2 = \{2, 4, 6, 8, 10, 9, 12, 15\}$
- Sorted in descending order: $\{15, 12, 10, 9, 8, 6, 4, 2\}$
- Rank 1: 15, Rank 2: 12, Rank 3: 10

Example 2: $a = 1, b = 5, c = 6, d = 10, k = 1$

- S_1 = even numbers in $[1, 5] = \{2, 4\}$
- S_2 = multiples of 3 in $[6, 10] = \{6, 9\}$
- $U = \{2, 4, 6, 9\}$
- Sorted in descending order: $\{9, 6, 4, 2\}$
- Rank 1: 9

Example 3: $a = 1, b = 1, c = 1, d = 1, k = 1$

- S_1 = even numbers in $[1, 1] = \{\}$ (1 is odd)
- S_2 = multiples of 3 in $[1, 1] = \{\}$ (1 is not a multiple of 3)
- $U = \{\}$ (empty set)
- No element with rank 1

Example 4: $a = 2, b = 100, c = 3, d = 100, k = 10$

- S_1 : even numbers in $[2, 100]$ (50 elements)
- S_2 : multiples of 3 in $[3, 100]$ (33 elements: 3,6,9,...,99)
- $S_1 \cap S_2$: multiples of 6 in $[\max(2, 3) = 3, 100]$ (16 elements: 6,12,...,96)
- $|U| = 50 + 33 - 16 = 67$
- The 10th largest element is 94

Constraints

- $1 \leq a \leq b \leq 10^9$
- $1 \leq c \leq d \leq 10^9$
- $1 \leq k \leq 10^{18}$